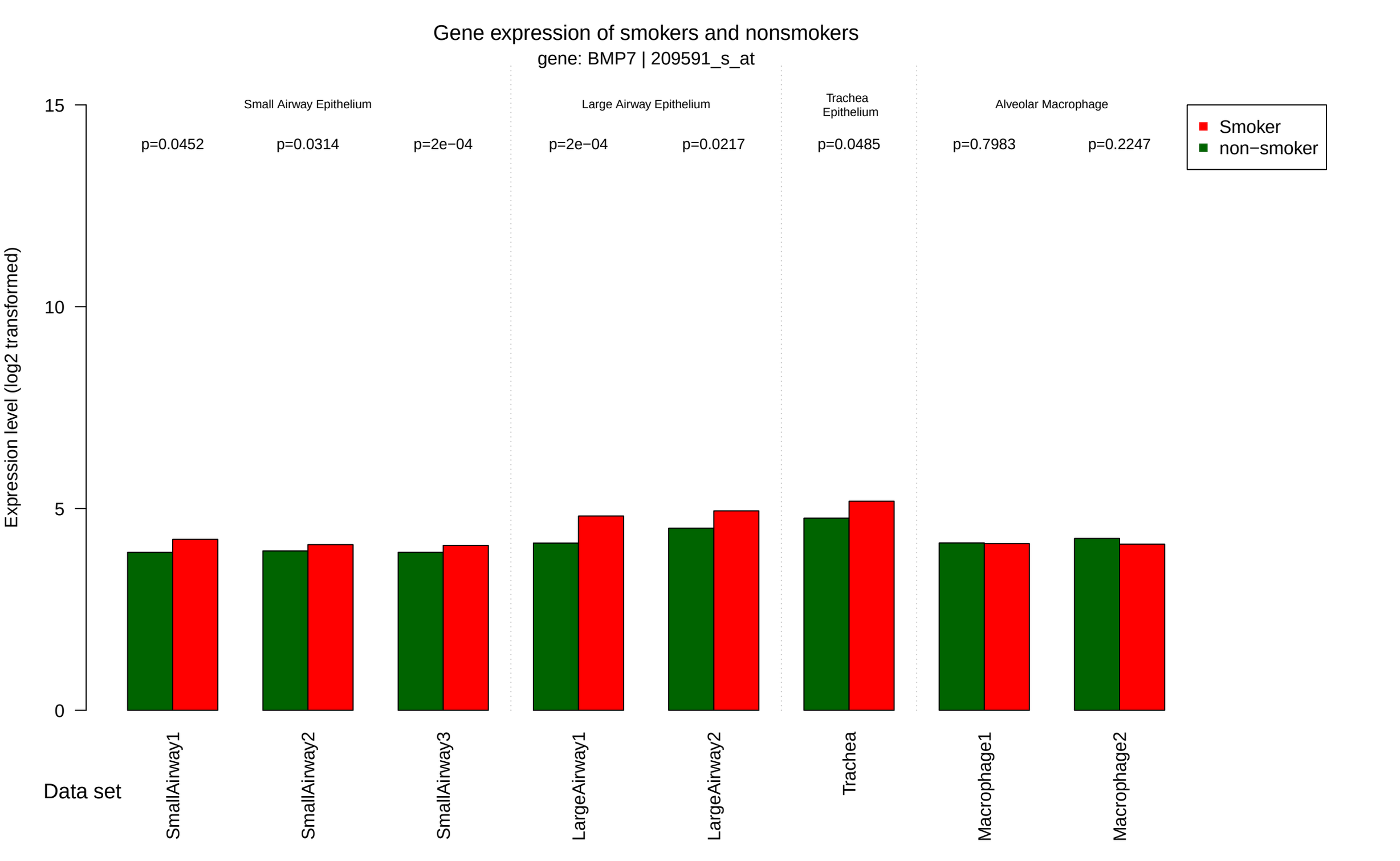
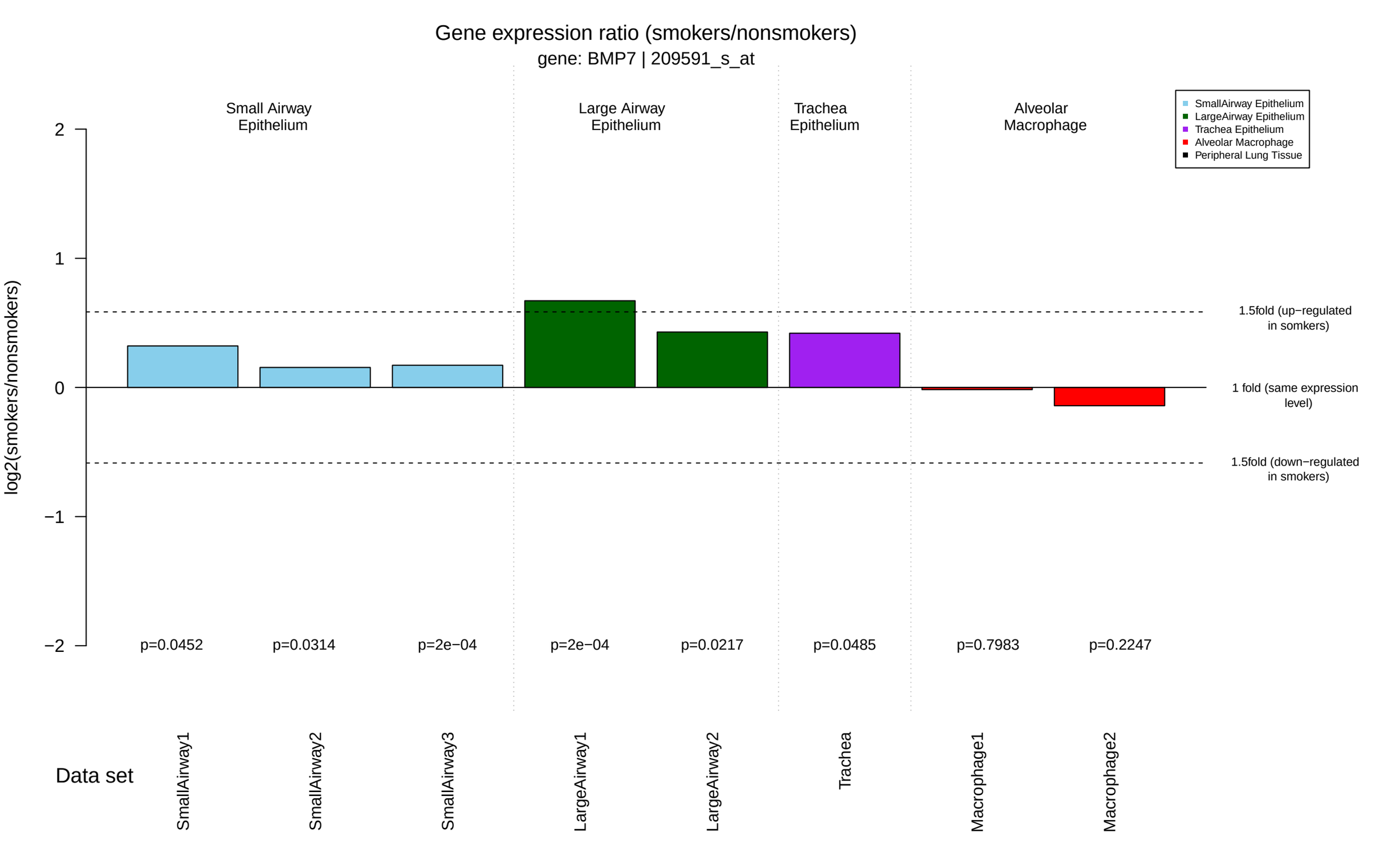
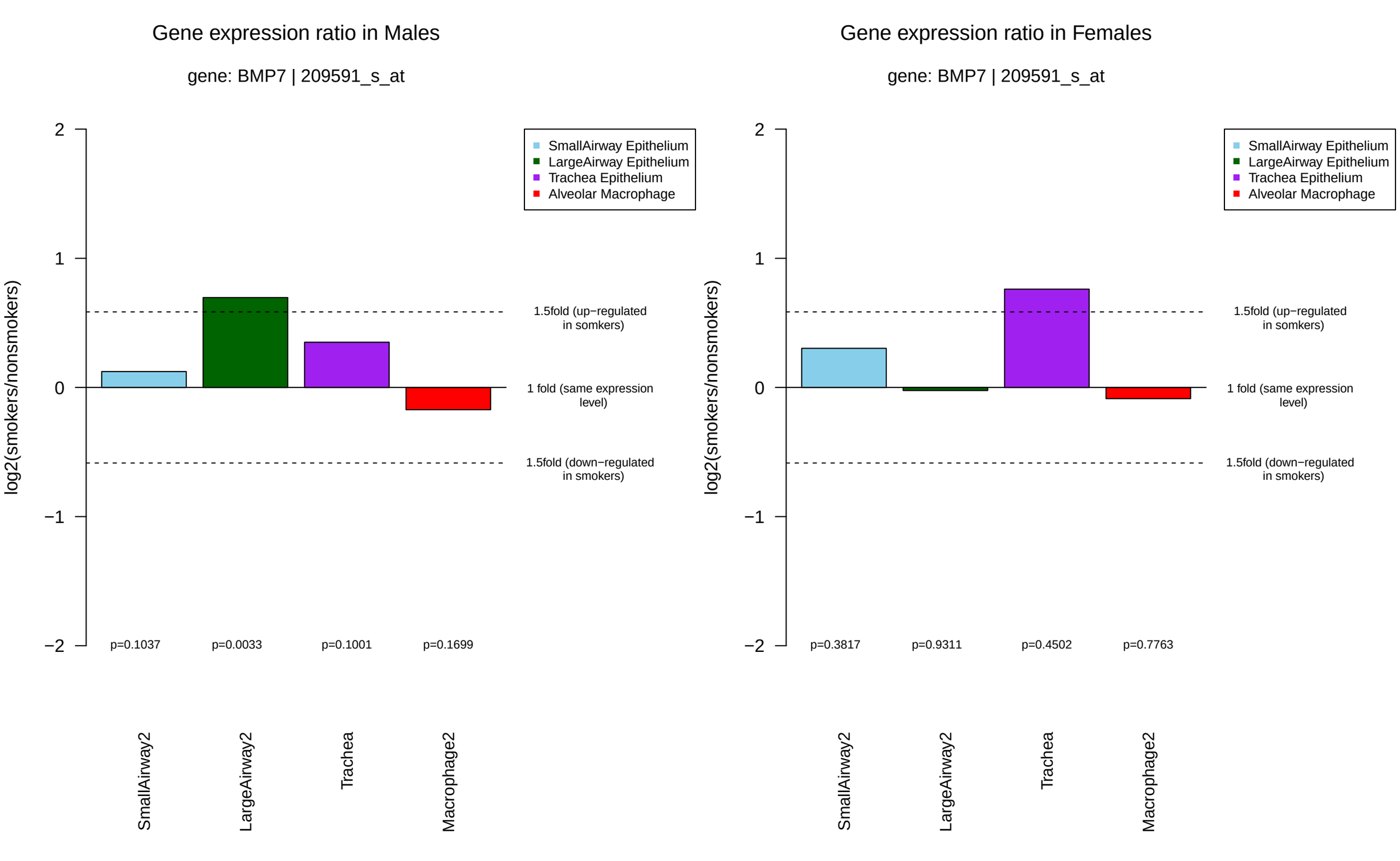
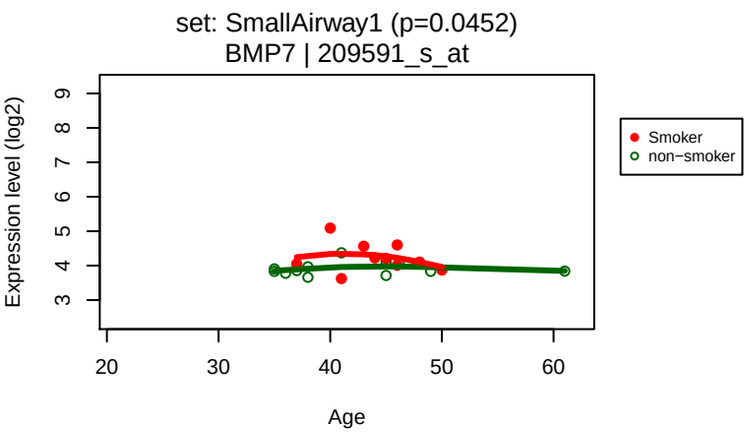
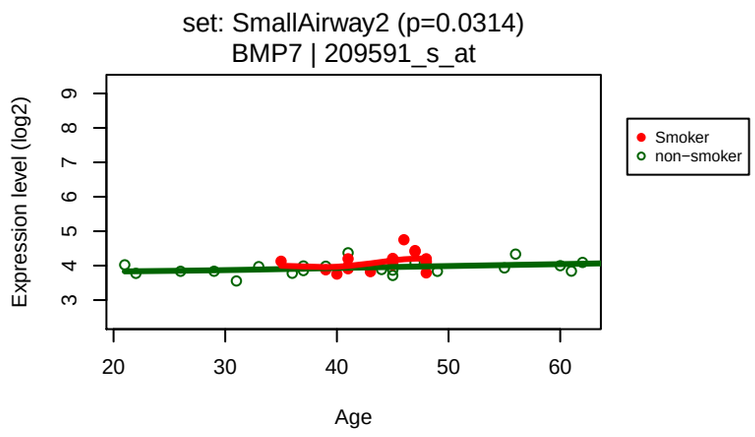
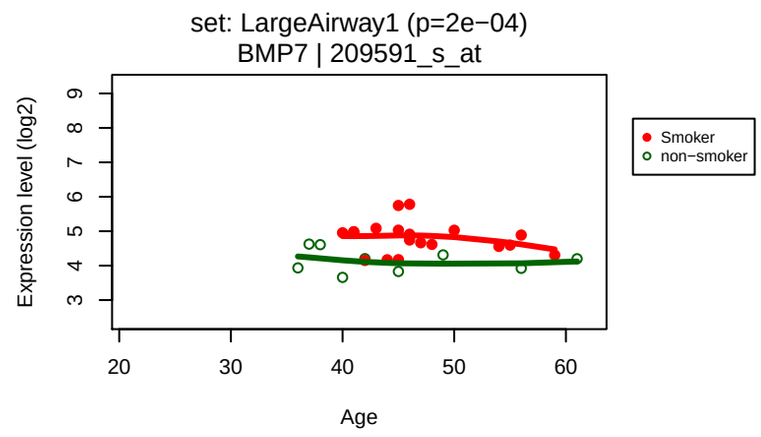

Gene expression of smokers and nonsmokers
gene: BMP7 | 209591_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0452
p=0.0314
p=2e−04
p=2e−04
p=0.0217
p=0.0485
p=0.7983
p=0.2247
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

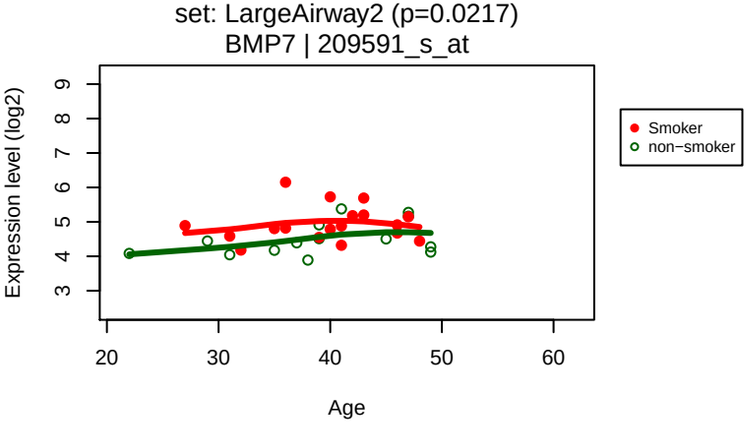

Gene expression ratio (smokers/nonsmokers)
gene: BMP7 | 209591_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0452
p=0.0314
p=2e−04
p=2e−04
p=0.0217
p=0.0485
p=0.7983
p=0.2247
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

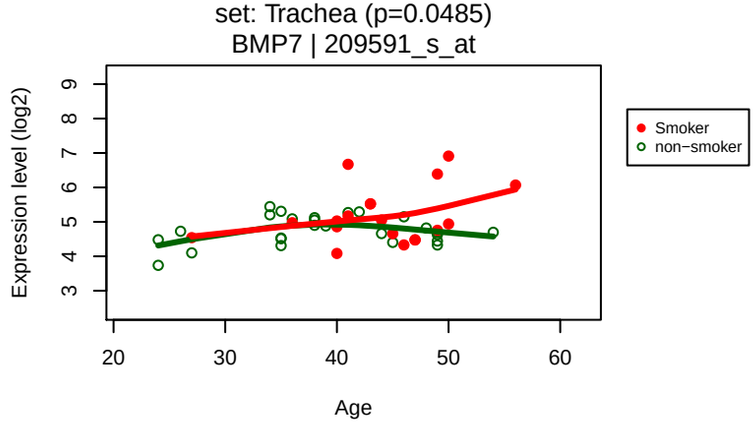

Gene expression ratio in Males
gene: BMP7 | 209591_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1037
p=0.0033
p=0.1001
p=0.1699
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: BMP7 | 209591_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3817
p=0.9311
p=0.4502
p=0.7763
SmallAirway2
LargeAirway2
Trachea
Macrophage2

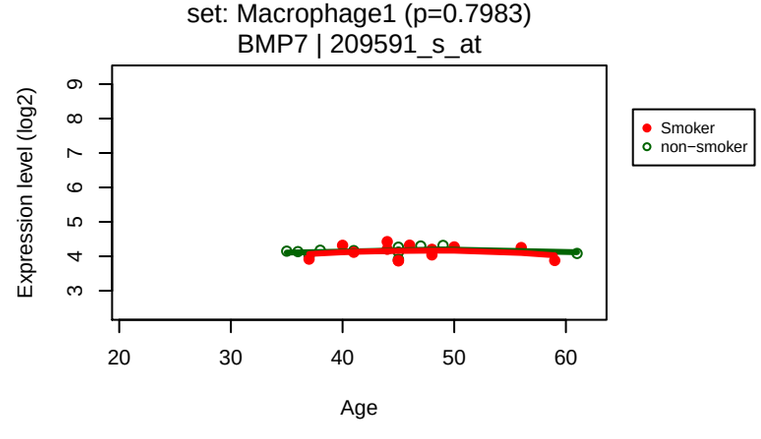

set: SmallAirway1 (p=0.0452)
BMP7 | 209591_s_at
Expression level (log2)
Age
Smoker
non−smoker

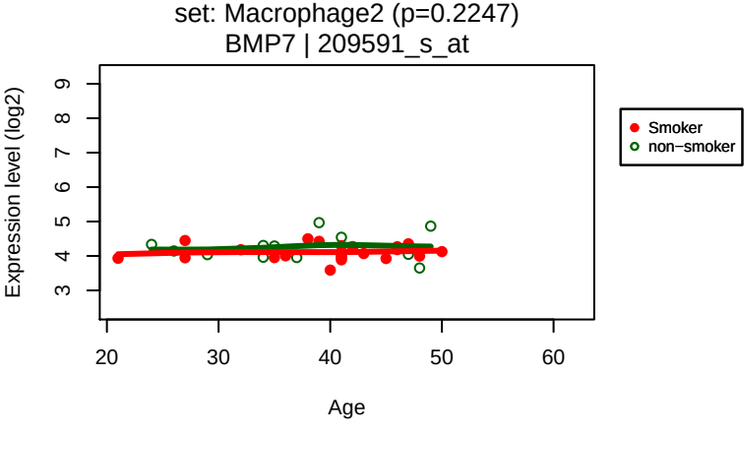

set: SmallAirway2 (p=0.0314)
BMP7 | 209591_s_at
Expression level (log2)
Age
Smoker
non−smoker

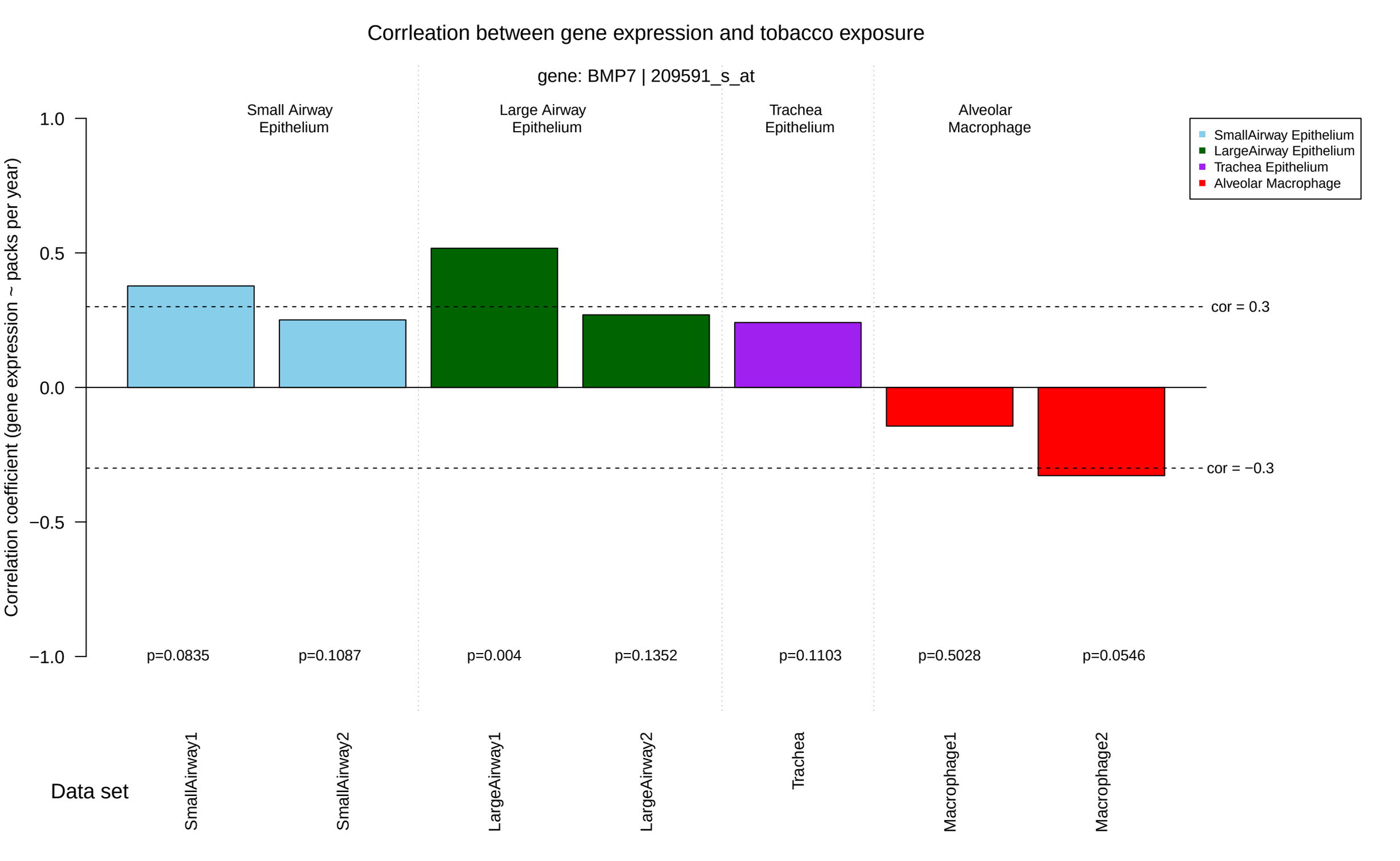

set: LargeAirway1 (p=2e−04)
BMP7 | 209591_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0217)
BMP7 | 209591_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0485)
BMP7 | 209591_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7983)
BMP7 | 209591_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2247)
BMP7 | 209591_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: BMP7 | 209591_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0835
p=0.1087
p=0.004
p=0.1352
p=0.1103
p=0.5028
p=0.0546
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2